

ICS344 Course Project Guide

Overview

The ICS344 course project involves setting up and attacking a vulnerable service, analyzing the attack with a SIEM (Security Information and Event Management) platform, and proposing a defensive strategy. The project is divided into three phases:

1. **Phase 1: Setup and Compromise the Service**
2. **Phase 2: Visual Analysis with a SIEM Dashboard**
3. **Phase 3: Defensive Strategy Proposal**



Groups

- ♦ **form a group of 3 from the same section**



Deadline May 3



Phase 1: Setup and Compromise the Service

In this phase, you will create two virtual environments:

1. **Victim Environment:** Configure a machine hosting the target service.
2. **Attacker Environment:** Use various tools to compromise the service.

Your **Victim machine** is **Metasploitable3**, which is a VM designed with numerous security vulnerabilities for testing exploits. Installation instructions can be found at the end of this file.

Vulnerable Services on Metasploitable3:

- ♦ IIS FTP

- ◆ IIS HTTP
- ◆ SSH
- ◆ WinRM
- ◆ WebDAV
- ◆ SNMP
- ◆ SMB
- ◆ Remote Desktop (RDP)

Metasploitable3 is intended for testing with [Metasploit](#).

Tasks for Phase 1

- ◆ **Task 1.1:** Use Kali Linux tool Metasploit to compromise the service.
- ◆ **Task 1.2:** Compromise the service using a custom script that you create (an automated script that compromises the target and shows a Proof of Concept).



Phase 2: Visual Analysis with a SIEM Dashboard

Use a SIEM platform to collect and visualize data from both the victim and honeypot environments.

Tasks for Phase 2

- ◆ Integrate logs from the victim environment into the SIEM platform.
- ◆ Visualize the attacks and compare the data from the victim.

Recommended SIEM Tool

- ◆ **Splunk**



Phase 3: Defensive Strategy Proposal

In this phase, propose a defense mechanism for the service:

- ◆ **You are allowed to implement any defensive mechanism to protect the service on the victim machine. You can use any tool that might help you achieve that.**

- ◆ **You must show how your defensive mechanism mitigates the attack. This means you need to rerun the attack after implementing the defense and prove it works.**
- ◆ **Demonstrate Before-and-After Security Status.**

Examples of Defense Mechanisms:

1. Apply Security Patches and Updates

- ◆ Manually patch vulnerabilities by downloading and applying specific patches to vulnerable services (e.g., SMB, RDP, FTP).

2. Modify Configuration Files

- ◆ Secure configuration settings (e.g., disable SMBv1, configure secure RDP connections, enforce password complexity).

3. Implement Security Tools

- ◆ Install security tools like firewalls, IDS/IPS, Fail2Ban directly on the VM.



Deliverables

- ◆ **GitHub Repository:** Students must submit their project as a GitHub repository containing all relevant setup configurations, attack scripts, analysis, and documentation. Ensure the repository is well-organized, clearly labeled, and publicly accessible.
 - ◆ include screen shots and a proof of everything that is mentioned in the Project Requirements Table.
 - ◆ **Create 1 folder for each phase.**
 - ◆ **You can create more subfolders if necessary.**
 - ◆ **Start working on it from the beginning. The grading will be based on your GitHub repository.**
 - ◆ Create a README file that contains
 - ◆ Group Number
 - ◆ work distribution
 - ◆ ID of each student
- ◆ **Presentation:** Teams should present key findings.
- ◆ For phase 1 submit your GitHub that contains phase 1 folder
- ◆ For phase 2 submit your GitHub that contains phase 1 folder and phase 2 folder
- ◆ For phase3 submit your GitHub that contains phase 1 folder and phase 2 folder and phase 3 folder

- ◆ Your grade will be the sum of your grades in each phase.



Project Requirements Table

Phase	Requirements
Phase 1: Setup and Compromise the Service	- Victim Environment: Set up Metasploitable3 with a vulnerable service.
	- Attacker Environment: Set up Kali Linux with Metasploit and other necessary tools.
	- Service Selection: Choose 1 service to attack from the list of vulnerable services.
	- Attack Execution: Use Metasploit and/or custom script to successfully compromise the selected service.
Deliverables:	
	Screenshot of steps compromising the service using Metasploit.
	- Screenshot of custom script.
	- Screenshot of service compromise using custom script.
Phase 2: SIEM Dashboard Analysis	- SIEM Setup: Integrate logs from both victim and attacker environments into a SIEM tool (e.g., Splunk).
	- Log Visualization: Visualize and analyze attack logs to understand attack patterns and behaviors.
	- Attack Comparison: Compare data from the victim and attacker environments.
Deliverables:	
	- Screenshot of SIEM integration (logs from both environments).
	- Screenshot of attack visualization and analysis.
Phase 3: Defensive Strategy Proposal	- Defense Mechanism: Propose and implement a defensive strategy (e.g., security patches, configuration changes).
	- Testing & Validation: Rerun the attack to validate the effectiveness of the defense mechanism.
	- Before-and-After Comparison: Show how the defense mitigates the attack by comparing the security

Phase	Requirements
	status before and after.
Deliverables:	
	- Screenshot of applied defense (e.g., patching, security tool installation).
	- Screenshot of before-and-after comparison showing security improvement.



Grading Criteria

- ◆ **Phase 1: Setup and Attacks** – 30%
- ◆ **Phase 2: SIEM Analysis** – 40%
- ◆ **Phase 3: Defense** – 30%



Official Repository for the ICS 344 Course Project Guide

- ◆ <https://github.com/osamacs7/344-Setup-Guide>



Support and Questions

For any questions or technical assistance, please reach out to the project TA:

- ◆ **OSAMA RAFAT ALKARNAWI** (s202183150@kfupm.edu.sa) via **TEAMS**



Phase 1: Setup and Compromise the Service – 30%

Criteria	Excellent (85-100%)	Needs Improvement (50-69%)	Unsatisfactory (0-49%)
Setup & Attack Execution	Fully set up victim and attacker environments; attack executed successfully.	Setup is partially done or attack is not fully successful.	Setup is incomplete; attack not executed or failed.



Phase 2: SIEM Dashboard Analysis – 40%

Criteria	Excellent 70-100%)	Needs Improvement (50-69%)	Unsatisfactory (0-49%)
Log Integration & Attack Visualization	Logs are fully integrated, and attack is well visualized.	Logs partially integrated, with missing information; attack visualization incomplete.	No integration or attack visualization.



Phase 3: Defensive Strategy Proposal – 30%

Criteria	Excellent (70-100%)	Needs Improvement (50-69%)	Unsatisfactory (0-49%)
Defense Mechanism & Testing	Effective defense mechanism applied; successfully tested.	Defense mechanism not fully effective or testing is unclear.	No effective defense mechanism or testing done.



Grading Scale:

- ◆ **A (85-100%):** Excellent work, meets or exceeds expectations.
- ◆ **B (70-84%):** Good work, minor issues.

- ◆ **C (50-69%):** Needs improvement, some major issues.
- ◆ **D (0-49%):** Significant issues or incomplete work.



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